# Ex1: Data Preprocessing and Cleaning

**Dataset:** Telco Customer Churn ( `Telco-Customer-Churn.csv` )

Upload `Telco-Customer-Churn.csv` to the Colab **Files** panel before running. If the file is not uploaded, the notebook downloads the same dataset automatically.

## Task 1: Import the required Python libraries

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
import warnings
warnings.filterwarnings('ignore')

sns.set_style('whitegrid')
pd.set_option('display.max_columns', None)
```

## Task 2: Load the dataset and display first 10 records, last 10 records, shape and column names

```
file_name = "Telco-Customer-Churn.csv"
if not os.path.exists(file_name):  # file not uploaded -> read the same dataset from GitHub
    file_name = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"

df = pd.read_csv(file_name)

print("First 10 records:")
display(df.head(10))
```

```
First 10 records:
```

| | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | InternetService |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 7590-VHVEG | Female | 0 | Yes | No | 1 | No | No phone service | DSL |
| **1** | 5575-GNVDE | Male | 0 | No | No | 34 | Yes | No | DSL |
| **2** | 3668-QPYBK | Male | 0 | No | No | 2 | Yes | No | DSL |
| **3** | 7795-CFOCW | Male | 0 | No | No | 45 | No | No phone service | DSL |
| **4** | 9237-HQITU | Female | 0 | No | No | 2 | Yes | No | Fiber optic |
| **5** | 9305-CDSKC | Female | 0 | No | No | 8 | Yes | Yes | Fiber optic |
| **6** | 1452-KIOVK | Male | 0 | No | Yes | 22 | Yes | Yes | Fiber optic |
| **7** | 6713-OKOMC | Female | 0 | No | No | 10 | No | No phone service | DSL |
| **8** | 7892-POOKP | Female | 0 | Yes | No | 28 | Yes | Yes | Fiber optic |
| **9** | 6388-TABGU | Male | 0 | No | Yes | 62 | Yes | No | DSL |

| | OnlineSecurity | OnlineBackup | DeviceProtection | TechSupport | StreamingTV | StreamingMovies | Contract | PaperlessBilling | PaymentMethod |
|---|---|---|---|---|---|---|---|---|---|
| **0** | No | Yes | No | No | No | No | Month-to-month | Yes | Electronic check |
| **1** | Yes | No | Yes | No | No | No | One year | No | Mailed check |
| **2** | Yes | Yes | No | No | No | No | Month-to-month | Yes | Mailed check |
| **3** | Yes | No | Yes | Yes | No | No | One year | No | Bank transfer (automatic) |
| **4** | No | No | No | No | No | No | Month-to-month | Yes | Electronic check |
| **5** | No | No | Yes | No | Yes | Yes | Month-to-month | Yes | Electronic check |
| **6** | No | Yes | No | No | Yes | No | Month-to-month | Yes | Credit card (automatic) |
| **7** | Yes | No | No | No | No | No | Month-to-month | No | Mailed check |
| **8** | No | No | Yes | Yes | Yes | Yes | Month-to-month | Yes | Electronic check |
| **9** | Yes | Yes | No | No | No | No | One year | No | Bank transfer (automatic) |

| | MonthlyCharges | TotalCharges | Churn |
|---|---|---|---|
| **0** | 29.85 | 29.85 | No |

| | MonthlyCharges | TotalCharges | Churn |
|---|---|---|---|
| **1** | 56.95 | 1889.5 | No |
| **2** | 53.85 | 108.15 | Yes |
| **3** | 42.30 | 1840.75 | No |
| **4** | 70.70 | 151.65 | Yes |
| **5** | 99.65 | 820.5 | Yes |
| **6** | 89.10 | 1949.4 | No |
| **7** | 29.75 | 301.9 | No |
| **8** | 104.80 | 3046.05 | Yes |
| **9** | 56.15 | 3487.95 | No |

```
print("Last 10 records:")
display(df.tail(10))
```

```
Last 10 records:
```

| | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | InternetService |
|---|---|---|---|---|---|---|---|---|---|
| **7033** | 9767-FFLEM | Male | 0 | No | No | 38 | Yes | No | Fiber optic |
| **7034** | 0639-TSIQW | Female | 0 | No | No | 67 | Yes | Yes | Fiber optic |
| **7035** | 8456-QDAVC | Male | 0 | No | No | 19 | Yes | No | Fiber optic |
| **7036** | 7750-EYXWZ | Female | 0 | No | No | 12 | No | No phone service | DSL |
| **7037** | 2569-WGERO | Female | 0 | No | No | 72 | Yes | No | No |
| **7038** | 6840-RESVB | Male | 0 | Yes | Yes | 24 | Yes | Yes | DSL |
| **7039** | 2234-XADUH | Female | 0 | Yes | Yes | 72 | Yes | Yes | Fiber optic |
| **7040** | 4801-JZAZL | Female | 0 | Yes | Yes | 11 | No | No phone service | DSL |
| **7041** | 8361-LTMKD | Male | 1 | Yes | No | 4 | Yes | Yes | Fiber optic |
| **7042** | 3186-AJIEK | Male | 0 | No | No | 66 | Yes | No | Fiber optic |

| | OnlineSecurity | OnlineBackup | DeviceProtection | TechSupport | StreamingTV | StreamingMovies | Contract | PaperlessBilling | PaymentMethod |
|---|---|---|---|---|---|---|---|---|---|
| **7033** | No | No | No | No | No | No | Month-to-month | Yes | Credit card (automatic) |
| **7034** | Yes | Yes | Yes | No | Yes | No | Month-to-month | Yes | Credit card (automatic) |
| **7035** | No | No | No | No | Yes | No | Month-to-month | Yes | Bank transfer (automatic) |
| **7036** | No | Yes | Yes | Yes | Yes | Yes | One year | No | Electronic check |
| **7037** | No internet service | No internet service | No internet service | No internet service | No internet service | No internet service | Two year | Yes | Bank transfer (automatic) |
| **7038** | Yes | No | Yes | Yes | Yes | Yes | One year | Yes | Mailed check |
| **7039** | No | Yes | Yes | No | Yes | Yes | One year | Yes | Credit card (automatic) |
| **7040** | Yes | No | No | No | No | No | Month-to-month | Yes | Electronic check |
| **7041** | No | No | No | No | No | No | Month-to-month | Yes | Mailed check |
| **7042** | Yes | No | Yes | Yes | Yes | Yes | Two year | Yes | Bank transfer (automatic) |

| | MonthlyCharges | TotalCharges | Churn |
|---|---|---|---|
| **7033** | 69.50 | 2625.25 | No |
| **7034** | 102.95 | 6886.25 | Yes |
| **7035** | 78.70 | 1495.1 | No |
| **7036** | 60.65 | 743.3 | No |
| **7037** | 21.15 | 1419.4 | No |
| **7038** | 84.80 | 1990.5 | No |
| **7039** | 103.20 | 7362.9 | No |
| **7040** | 29.60 | 346.45 | No |
| **7041** | 74.40 | 306.6 | Yes |
| **7042** | 105.65 | 6844.5 | No |

```
print("Shape of the dataset:", df.shape)
print("Number of rows   :", df.shape[0])
print("Number of columns:", df.shape[1])
print("\nColumn names:")
print(df.columns.tolist())
```

```
Shape of the dataset: (7043, 21)
Number of rows   : 7043
```

```
Number of columns: 21

Column names:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']
```

## Task 3: Explore the dataset – information, summary statistics, missing values, duplicate records

```
print("Dataset information:")
df.info()
```

```
Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   str
 20  Churn             7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

```
print("Summary statistics (numerical columns):")
display(df.describe().T)

print("Summary statistics (categorical columns):")
display(df.describe(exclude='number').T)
```

Summary statistics (numerical columns):

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **SeniorCitizen** | 7043.0 | 0.162147 | 0.368612 | 0.00 | 0.0 | 0.00 | 0.00 | 1.00 |
| **tenure** | 7043.0 | 32.371149 | 24.559481 | 0.00 | 9.0 | 29.00 | 55.00 | 72.00 |
| **MonthlyCharges** | 7043.0 | 64.761692 | 30.090047 | 18.25 | 35.5 | 70.35 | 89.85 | 118.75 |

Summary statistics (categorical columns):

| | count | unique | top | freq |
|---|---|---|---|---|
| **customerID** | 7043 | 7043 | 7590-VHVEG | 1 |
| **gender** | 7043 | 2 | Male | 3555 |
| **Partner** | 7043 | 2 | No | 3641 |
| **Dependents** | 7043 | 2 | No | 4933 |
| **PhoneService** | 7043 | 2 | Yes | 6361 |
| **MultipleLines** | 7043 | 3 | No | 3390 |
| **InternetService** | 7043 | 3 | Fiber optic | 3096 |
| **OnlineSecurity** | 7043 | 3 | No | 3498 |
| **OnlineBackup** | 7043 | 3 | No | 3088 |
| **DeviceProtection** | 7043 | 3 | No | 3095 |
| **TechSupport** | 7043 | 3 | No | 3473 |
| **StreamingTV** | 7043 | 3 | No | 2810 |
| **StreamingMovies** | 7043 | 3 | No | 2785 |
| **Contract** | 7043 | 3 | Month-to-month | 3875 |
| **PaperlessBilling** | 7043 | 2 | Yes | 4171 |
| **PaymentMethod** | 7043 | 4 | Electronic check | 2365 |
| **TotalCharges** | 7043 | 6531 | 20.2 | 11 |
| **Churn** | 7043 | 2 | No | 5174 |

**Inconsistency found:** `TotalCharges` is stored as text (object) and contains blank strings `" "` instead of numbers. These blanks are hidden missing values, so the column is converted to numeric and the blanks become `NaN`.

```
print("Blank strings in TotalCharges:", (df['TotalCharges'].astype(str).str.strip() == '').sum())

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
print("TotalCharges data type after conversion:", df['TotalCharges'].dtype)
```

```
Blank strings in TotalCharges: 11
TotalCharges data type after conversion: float64
```

```
print("Number of missing values in each column:")
print(df.isnull().sum())
print("\nTotal missing values:", df.isnull().sum().sum())

print("\nNumber of duplicate records:", df.duplicated().sum())
print("Number of duplicate customer IDs:", df['customerID'].duplicated().sum())
```

```
Number of missing values in each column:
customerID           0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges        11
Churn                0
dtype: int64

Total missing values: 11

Number of duplicate records: 0
Number of duplicate customer IDs: 0
```

# Task 4: Clean the data

## 4a. Remove duplicate customer records

```
rows_before = df.shape[0]
df = df.drop_duplicates()                                    # exact duplicate rows
df = df.drop_duplicates(subset='customerID', keep='first')   # same customer repeated
print("Duplicate records removed:", rows_before - df.shape[0])
print("Shape after removing duplicates:", df.shape)
```

```
Duplicate records removed: 0
Shape after removing duplicates: (7043, 21)
```

## 4b. Replace missing values (Numerical → Median, Categorical → Mode)

```
num_cols = df.select_dtypes(include='number').columns.tolist()
cat_cols = df.select_dtypes(exclude='number').columns.tolist()
print("Numerical columns  :", num_cols)
print("Categorical columns:", cat_cols)

for col in num_cols:
    if df[col].isnull().sum() > 0:
        median_value = df[col].median()
        df[col] = df[col].fillna(median_value)
        print(f"{col}: missing values replaced with median = {median_value}")

for col in cat_cols:
    if df[col].isnull().sum() > 0:
        mode_value = df[col].mode()[0]
        df[col] = df[col].fillna(mode_value)
        print(f"{col}: missing values replaced with mode = {mode_value}")
```

```
Numerical columns  : ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']
Categorical columns: ['customerID', 'gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService',
'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
'PaperlessBilling', 'PaymentMethod', 'Churn']
TotalCharges: missing values replaced with median = 1397.475
```

### 4c. Verify that no missing values remain

```
print(df.isnull().sum())
print("\nTotal missing values after cleaning:", df.isnull().sum().sum())
```

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Total missing values after cleaning: 0
```

## Task 5: Encode categorical columns – Gender, Partner, Dependents, Contract, PaymentMethod, InternetService, Churn

- **Label Encoding** for binary columns (2 categories): `gender`, `Partner`, `Dependents`, `Churn`
- **One-Hot Encoding** for nominal columns with more than 2 categories: `Contract`, `PaymentMethod`, `InternetService`

```
df_clean = df.copy()      # cleaned data with original labels (used for the plots)
df_encoded = df.copy()    # data that will be encoded and scaled

binary_cols = ['gender', 'Partner', 'Dependents', 'Churn']
le = LabelEncoder()
for col in binary_cols:
    df_encoded[col] = le.fit_transform(df_encoded[col])
    print(f"{col}: {dict((c, i) for i, c in enumerate(le.classes_))}")

multi_cols = ['Contract', 'PaymentMethod', 'InternetService']
df_encoded = pd.get_dummies(df_encoded, columns=multi_cols, dtype=int)

print("\nShape after encoding:", df_encoded.shape)
display(df_encoded.head())
```

```
gender: {'Female': 0, 'Male': 1}
Partner: {'No': 0, 'Yes': 1}
Dependents: {'No': 0, 'Yes': 1}
Churn: {'No': 0, 'Yes': 1}

Shape after encoding: (7043, 28)
```

| | customerID | gender | SeniorCitizen | Partner | Dependents | tenure | PhoneService | MultipleLines | OnlineSecurity |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 7590-VHVEG | 0 | 0 | 1 | 0 | 1 | No | No phone service | No |
| **1** | 5575-GNVDE | 1 | 0 | 0 | 0 | 34 | Yes | No | Yes |
| **2** | 3668-QPYBK | 1 | 0 | 0 | 0 | 2 | Yes | No | Yes |
| **3** | 7795-CFOCW | 1 | 0 | 0 | 0 | 45 | No | No phone service | Yes |
| **4** | 9237-HQITU | 0 | 0 | 0 | 0 | 2 | Yes | No | No |

| | OnlineBackup | DeviceProtection | TechSupport | StreamingTV | StreamingMovies | PaperlessBilling | MonthlyCharges | TotalCharges | Churn |
|---|---|---|---|---|---|---|---|---|---|
| **0** | Yes | No | No | No | No | Yes | 29.85 | 29.85 | 0 |
| **1** | No | Yes | No | No | No | No | 56.95 | 1889.50 | 0 |
| **2** | Yes | No | No | No | No | Yes | 53.85 | 108.15 | 1 |
| **3** | No | Yes | Yes | No | No | No | 42.30 | 1840.75 | 0 |
| **4** | No | No | No | No | No | Yes | 70.70 | 151.65 | 1 |

| | Contract_Month-to-month | Contract_One year | Contract_Two year | PaymentMethod_Bank transfer (automatic) | PaymentMethod_Credit card (automatic) | PaymentMethod_Electronic check | PaymentMethod_Mailed check | InternetService_DSL | InternetService_Fiber optic |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| **1** | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| **2** | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |

| | Contract_Month-to-month | Contract_One year | Contract_Two year | PaymentMethod_Bank transfer (automatic) | PaymentMethod_Credit card (automatic) | PaymentMethod_Electronic check | PaymentMethod_Mailed check | InternetService_DSL | InternetService_Fiber optic |
|---|---|---|---|---|---|---|---|---|---|
| **3** | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| **4** | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |

| | InternetService_No |
|---|---|
| **0** | 0 |
| **1** | 0 |
| **2** | 0 |
| **3** | 0 |
| **4** | 0 |

## Task 6: Standardise the numerical features – Tenure, MonthlyCharges, TotalCharges (StandardScaler)

```
scale_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']

print("Before scaling:")
display(df_encoded[scale_cols].describe().round(2))

scaler = StandardScaler()
df_encoded[scale_cols] = scaler.fit_transform(df_encoded[scale_cols])

print("After scaling (mean ≈ 0, std ≈ 1):")
display(df_encoded[scale_cols].describe().round(2))
```

```
Before scaling:
```

| | tenure | MonthlyCharges | TotalCharges |
|---|---|---|---|
| **count** | 7043.00 | 7043.00 | 7043.00 |
| **mean** | 32.37 | 64.76 | 2281.92 |
| **std** | 24.56 | 30.09 | 2265.27 |
| **min** | 0.00 | 18.25 | 18.80 |
| **25%** | 9.00 | 35.50 | 402.22 |
| **50%** | 29.00 | 70.35 | 1397.48 |
| **75%** | 55.00 | 89.85 | 3786.60 |
| **max** | 72.00 | 118.75 | 8684.80 |

```
After scaling (mean ≈ 0, std ≈ 1):
```

| | tenure | MonthlyCharges | TotalCharges |
|---|---|---|---|
| **count** | 7043.00 | 7043.00 | 7043.00 |
| **mean** | -0.00 | -0.00 | -0.00 |
| **std** | 1.00 | 1.00 | 1.00 |
| **min** | -1.32 | -1.55 | -1.00 |
| **25%** | -0.95 | -0.97 | -0.83 |
| **50%** | -0.14 | 0.19 | -0.39 |
| **75%** | 0.92 | 0.83 | 0.66 |
| **max** | 1.61 | 1.79 | 2.83 |

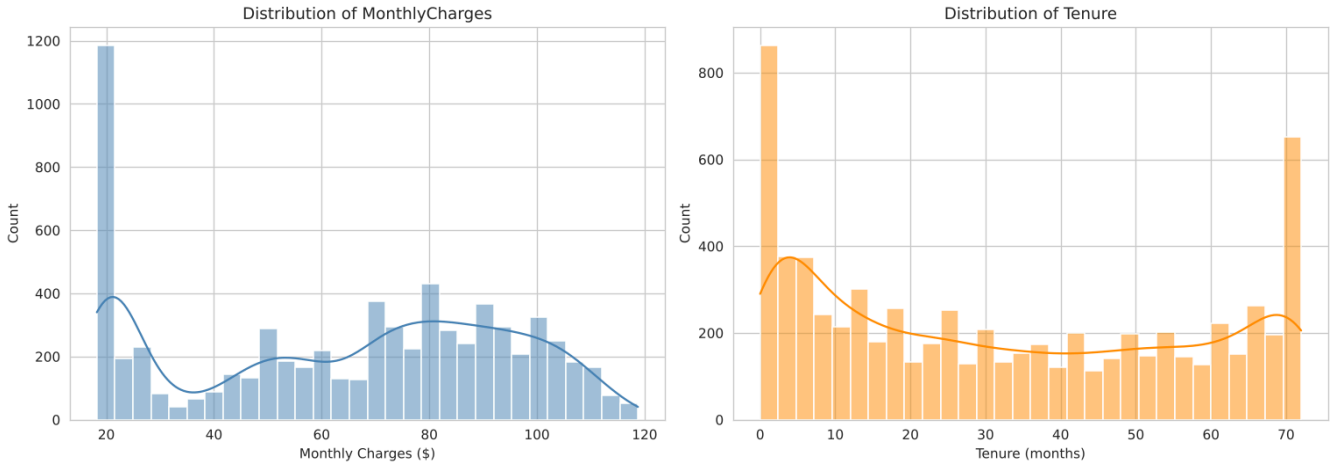
## Task 7: Plot the distribution of MonthlyCharges and Tenure

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.histplot(df_clean['MonthlyCharges'], bins=30, kde=True, color='steelblue', ax=axes[0])
axes[0].set_title('Distribution of MonthlyCharges')
axes[0].set_xlabel('Monthly Charges ($)')

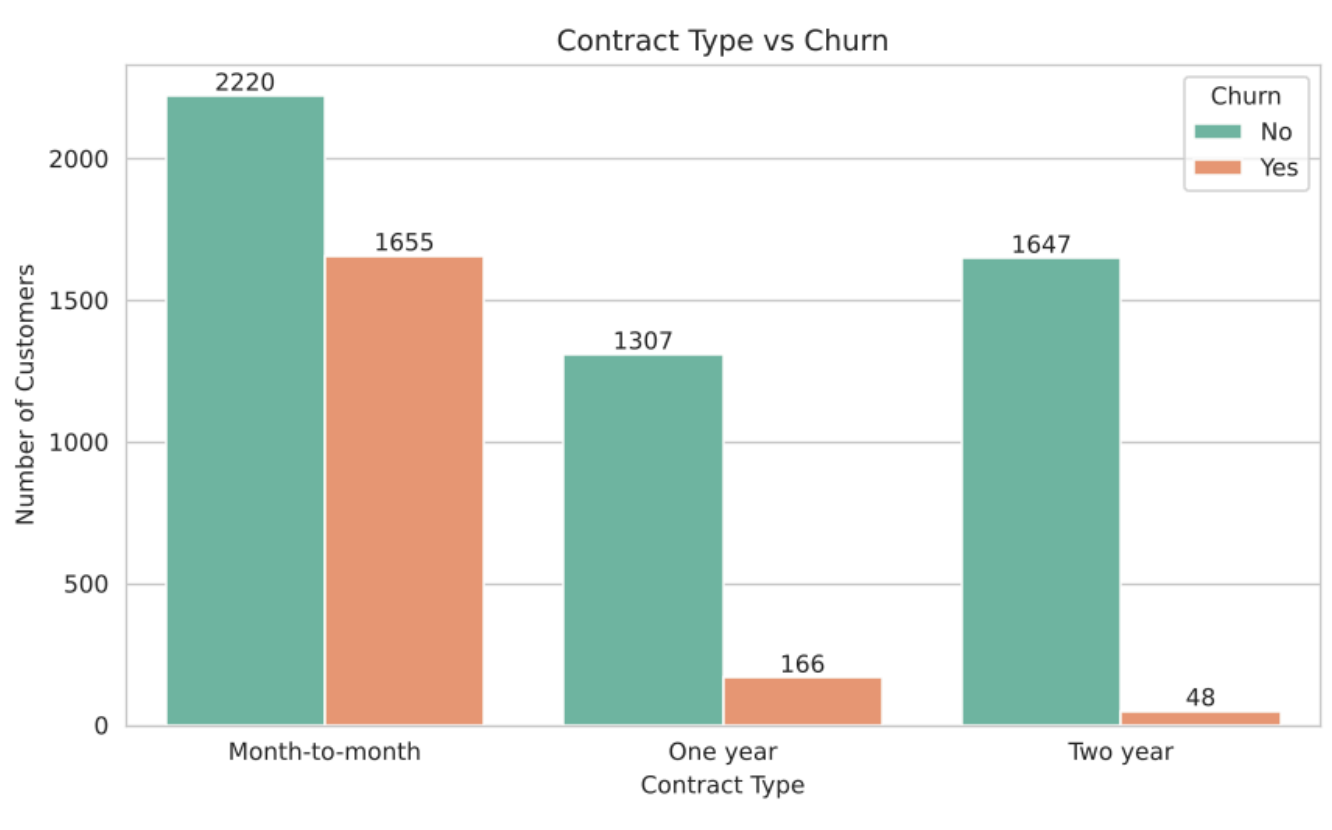
sns.histplot(df_clean['tenure'], bins=30, kde=True, color='darkorange', ax=axes[1])
axes[1].set_title('Distribution of Tenure')
axes[1].set_xlabel('Tenure (months)')

plt.tight_layout()
plt.show()
```



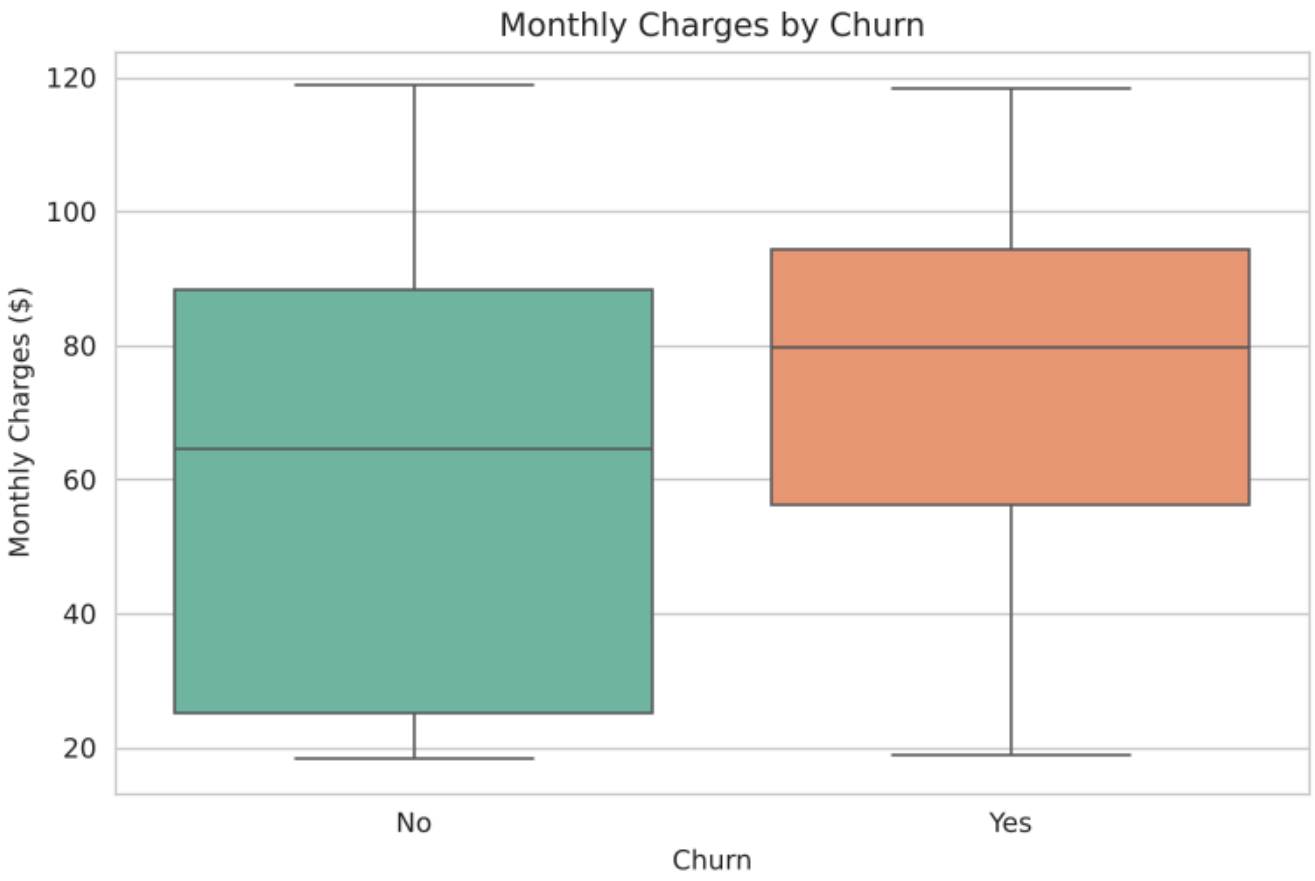

## Task 8: Count Plot – Contract vs Churn

```
plt.figure(figsize=(9, 5))
ax = sns.countplot(data=df_clean, x='Contract', hue='Churn', palette='Set2')
for container in ax.containers:
    ax.bar_label(container)
plt.title('Contract Type vs Churn')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.show()
```




## Task 9: Box Plot – MonthlyCharges grouped by Churn

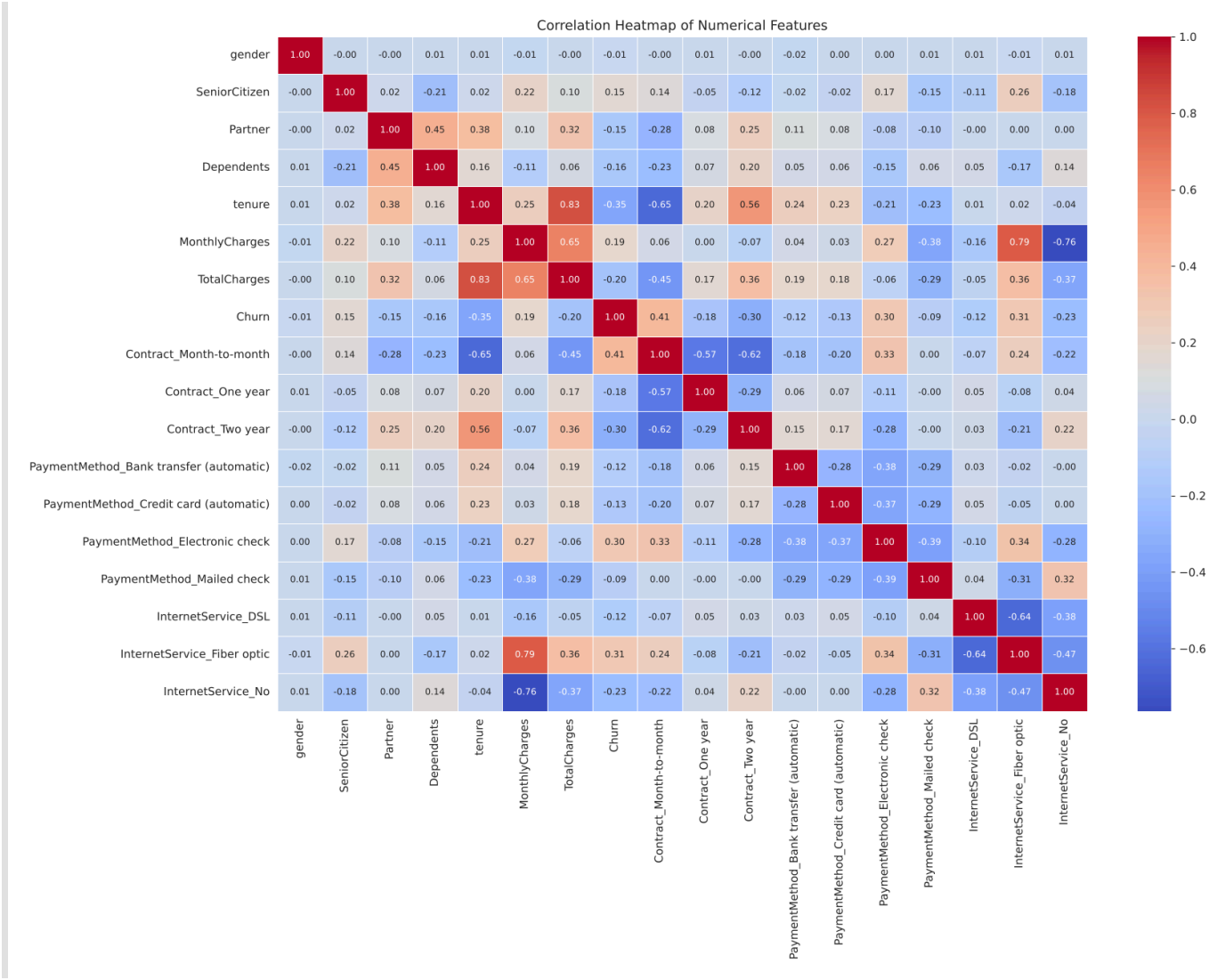
```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df_clean, x='Churn', y='MonthlyCharges', hue='Churn', palette='Set2', legend=False)
plt.title('Monthly Charges by Churn')
plt.xlabel('Churn')
plt.ylabel('Monthly Charges ($)')
plt.show()
```




## Task 10: Correlation Heatmap of all numerical features

```
corr = df_encoded.select_dtypes(include='number').corr()

plt.figure(figsize=(16, 12))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', linewidths=0.5, annot_kws={'size': 8})
plt.title('Correlation Heatmap of Numerical Features')
plt.tight_layout()
plt.show()
```


Correlation Heatmap of Numerical Features
gender 1.00 -0.00 -0.00 0.01 0.01 -0.01 -0.00 -0.01 -0.00 0.01 -0.00 -0.02 0.00 0.00 0.01 0.01 -0.01 0.01
SeniorCitizen -0.00 1.00 0.02 -0.21 0.02 0.22 0.10 0.15 0.14 -0.05 -0.12 -0.02 -0.02 0.17 -0.15 -0.11 0.26 -0.18
Partner -0.00 0.02 1.00 0.45 0.38 0.10 0.32 -0.15 -0.28 0.08 0.25 0.11 0.08 -0.08 -0.10 -0.00 0.00 0.00
Dependents 0.01 -0.21 0.45 1.00 0.16 -0.11 0.06 -0.16 -0.23 0.07 0.20 0.05 0.06 -0.15 0.06 0.05 -0.17 0.14
tenure 0.01 0.02 0.38 0.16 1.00 0.25 0.83 -0.35 -0.65 0.20 0.56 0.24 0.23 -0.21 -0.23 0.01 0.02 -0.04
MonthlyCharges -0.01 0.22 0.10 -0.11 0.25 1.00 0.65 0.19 0.06 0.00 -0.07 0.04 0.03 0.27 -0.38 -0.16 0.79 -0.76
TotalCharges -0.00 0.10 0.32 0.06 0.83 0.65 1.00 -0.20 -0.45 0.17 0.36 0.19 0.18 -0.06 -0.29 -0.05 0.36 -0.37
Churn -0.01 0.15 -0.15 -0.16 -0.35 0.19 -0.20 1.00 0.41 -0.18 -0.30 -0.12 -0.13 0.30 -0.09 -0.12 0.31 -0.23
Contract_Month-to-month -0.00 0.14 -0.28 -0.23 -0.65 0.06 -0.45 0.41 1.00 -0.57 -0.62 -0.18 -0.20 0.33 0.00 -0.07 0.24 -0.22
Contract_One year 0.01 -0.05 0.08 0.07 0.20 0.00 0.17 -0.18 -0.57 1.00 -0.29 0.06 0.07 -0.11 -0.00 0.05 -0.08 0.04
Contract_Two year -0.00 -0.12 0.25 0.20 0.56 -0.07 0.36 -0.30 -0.62 -0.29 1.00 0.15 0.17 -0.28 -0.00 0.03 -0.21 0.22
PaymentMethod_Bank transfer (automatic) -0.02 -0.02 0.11 0.05 0.24 0.04 0.19 -0.12 -0.18 0.06 0.15 1.00 -0.28 -0.38 -0.29 0.03 -0.02 -0.00
PaymentMethod_Credit card (automatic) 0.00 -0.02 0.08 0.06 0.23 0.03 0.18 -0.13 -0.20 0.07 0.17 -0.28 1.00 -0.37 -0.29 0.05 -0.05 0.00
PaymentMethod_Electronic check 0.00 0.17 -0.08 -0.15 -0.21 0.27 -0.06 0.30 0.33 -0.11 -0.28 -0.38 -0.37 1.00 -0.39 -0.10 0.34 -0.28
PaymentMethod_Mailed check 0.01 -0.15 -0.10 0.06 -0.23 -0.38 -0.29 -0.09 0.00 -0.00 -0.00 -0.29 -0.29 -0.39 1.00 0.04 -0.31 0.32
InternetService_DSL 0.01 -0.11 -0.00 0.05 0.01 -0.16 -0.05 -0.12 -0.07 0.05 0.03 0.03 0.05 -0.10 0.04 1.00 -0.64 -0.38
InternetService_Fiber optic -0.01 0.26 0.00 -0.17 0.02 0.79 0.36 0.31 0.24 -0.08 -0.21 -0.02 -0.05 0.34 -0.31 -0.64 1.00 -0.47
InternetService_No 0.01 -0.18 0.00 0.14 -0.04 -0.76 -0.37 -0.23 -0.22 0.04 0.22 -0.00 0.00 -0.28 0.32 -0.38 -0.47 1.00
gender
SeniorCitizen
Partner
Dependents
tenure
MonthlyCharges
TotalCharges
Churn
Contract_Month-to-month
Contract_One year
Contract_Two year
PaymentMethod_Bank transfer (automatic)
PaymentMethod_Credit card (automatic)
PaymentMethod_Electronic check
PaymentMethod_Mailed check
InternetService_DSL
InternetService_Fiber optic
InternetService_No
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6